ANL Staking Protocol – Audit #5 (Static Review)

Executive Summary

The reviewed changes split the previous single initialize instruction into four separate initialization instructions in order to eliminate the Solana SBF stack overflow problem. Based on static inspection, the protocol logic, checkpoint model and vault ownership invariants remain consistent.

Findings

- No critical vulnerabilities identified.
- No high-severity vulnerabilities identified.
- Intermediate state between initialize and init_*_vault appears protected by authority validation, PDA-derived vault addresses and mandatory account initialization checks.
- Box / Box changes reduce stack usage without changing Anchor constraints.
- Migration to Anchor 0.30.1 appears internally consistent.
- Separate Program IDs per network are a positive hardening measure.
- API changed because initialization is now multi-stage; clients must be updated.

Limitations

This assessment is a static audit only. The environment did not contain cargo, Anchor or Solana tooling, therefore compilation, on-chain deployment and test execution could not be independently reproduced.

Verdict

Closed Testnet: READY after successful execution of the full integration suite on the real toolchain.

Immutable Mainnet: NOT YET READY. Before deployment, complete a final re-validation of the new initialization architecture, verify the deployable SBF artifact, ensure configuration consistency and complete the production release checklist.